

Polynomials

** Full solutions are required for full marks **

Total
35 / 36

1. For a polynomial function with zeros 5; 2 (order 2); and -6 (order 3) passes through the point (1, -343), determine:

a) an equation to represent the function: (1.5)

$$y = a(x-5)(x-2)^2(x+6)^3$$

$$-343 = a(1-5)(1-2)^2(1+6)^3$$

$$-343 = -4(343)a$$

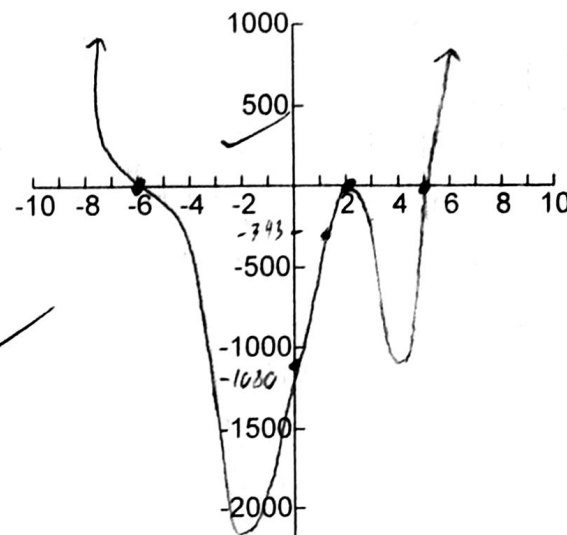
$$a = \frac{1}{4}$$

$$y = \frac{1}{4}(x-5)(x-2)^2(x+6)^3$$

b) degree of the function 6 (5)

c) y-intercept (0, -1080) (1)

d) a graph for the function (show all intercepts) (1)



2. a) Sketch a graph, using mapping rule, of the function $f(x) = x^4$ that has been transformed by

$y = -2f\left(\frac{1}{3}x - \frac{2}{3}\right) + 7$. Sketch the transformed function. Remember to label your graph.

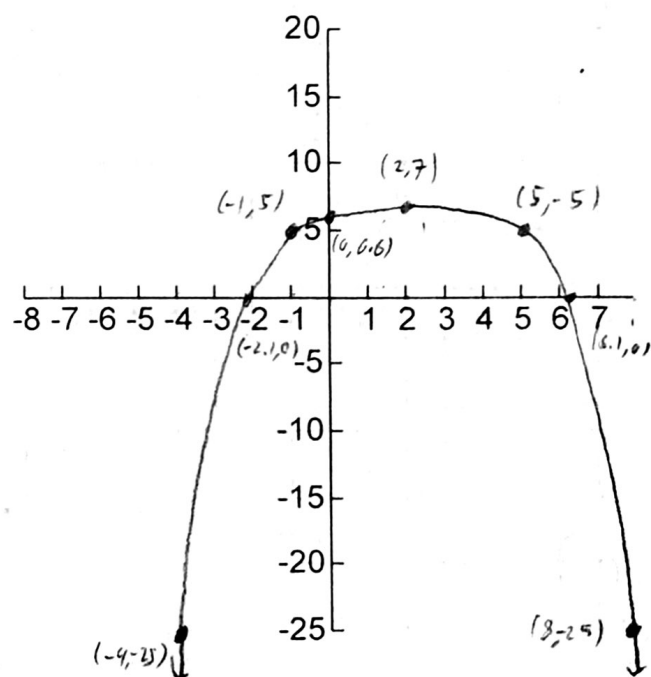
$$\frac{1}{3}(x-2) \quad a=-2, k=\frac{1}{3}, d=2, c=7$$

$$f(x) = x^4$$

$$y = -2\left[\frac{1}{3}(x-2)\right]^4 + 7$$

x	y
-2	16
-1	1
0	0
1	1
2	16

3x+2	-2y+7
-4	-25
-1	5
2	7
5	5
8	-25



b) Describe the transformations on $f(x) = x^4$.

• Reflection on the x-axis ✓

• Vertical stretch by a factor of 2 ✓

• Horizontal stretch by a factor of 3 ✓

• Horizontal shift Right by 2 units ✓

• Vertical Shift Up by 7 units ✓

$$y = -2\left[\frac{1}{3}(x-2)\right]^4 + 7$$

3. Determine the Instantaneous Rate of Change at 4 seconds. Round to 1 d.p.

$$h(t) = a(t+3)(t-2)^2(t-4)$$

$$-144 = a(4+3)(4-2)^2(4-4)$$

$$-144 = -48a$$

$$a = 3$$

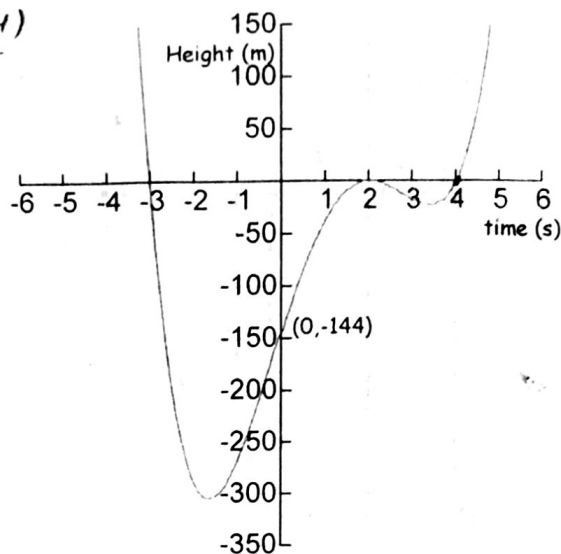
$$(3) \quad h(t) = 3(t+3)(t-2)^2(t-4)$$

$$h(4) = 3(4+3)(4-2)^2(4-4) = 0$$

$$h(4.001) = 3(4.001+3)(4.001-2)^2(4.001-4) = 0.084096033$$

$$I.R.C. = \frac{h(4.001) - h(4)}{4.001 - 4}$$

$$= 84.1 \text{ m/s}$$



4. Show if the function $f(x) = -2(x-2)^3(x+2)^3$ is odd, even or neither algebraically.

$$f(-x) = -2(-x-2)^3(-x+2)^3$$

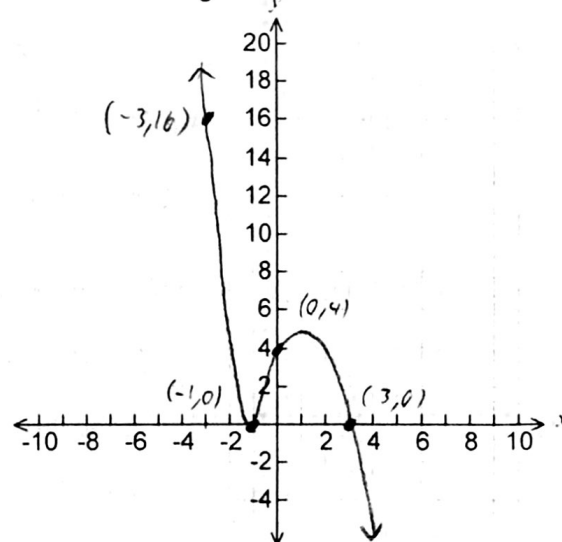
$$= -2(x+2)^3(x-2)^3$$

$$f(-x) = f(x)$$

$\therefore$ The function is even

5. Draw a graph of a polynomial function that satisfies all of the following characteristics:

- $f(-3) = 16$, $f(3) = 0$ and $f(-1) = 0$
- y-intercept is 4
- $f(x) \geq 0$ when $x < 3$
- The domain is the set of real numbers.



6. A section of a roller coaster is modeled by the function $h(t) = -(t-1)^3 + 0.5(t-1)^2$, where h is the depth in meters after t seconds. Round to 2 d.p.

a) Determine the average rate of change from 4 to 6 seconds.

(2) $(4, -22.5)$
 $(6, -112.5)$ $AROC = \frac{-112.5 - (-22.5)}{6-4} = -45 \text{ m/s}$ ✓

b) Determine the velocity of the coaster at 6 seconds. ✓

(2) $(6, -112.5)$
 $(6.001, -122.5700145)$ $\vec{V} = IROC = \frac{-122.5700145 - (-112.5)}{6.001 - 6} = -70.01 \text{ m/s}$
 $= 70.01 \text{ m/s [down]}$

c) Using your answers from part a) and b), describe the speed and the direction of the roller coaster as it approaches 6 seconds.

✓ As t approaches 6 seconds, the speed approaches 70 m/s and the direction of the roller coaster is heading downwards.

7. Given a polynomial function $f(x)$ of degree n , where n is a positive even integer, determine whether each statement is true or false. If the statement is false, give an example that illustrates why it is false. ✓

i. $f(x)$ is an even function

T or (F) $f(x) = x^4 + x^3$ is not an even function. ✓

ii. $f(x)$ will have at least one zero

T or (F) $f(x) = x^2 + 1$ is above the x -axis ✓

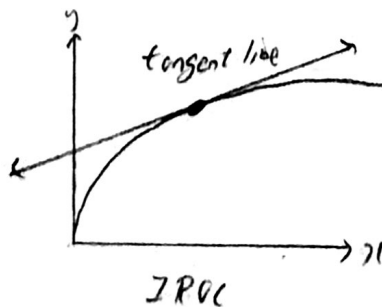
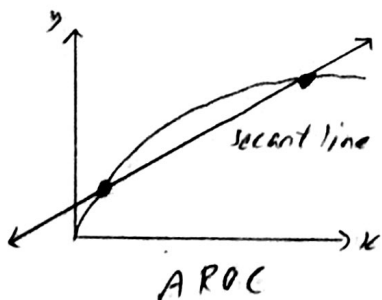
iii. as $x \rightarrow \infty, y \rightarrow \infty$ and as $x \rightarrow -\infty, y \rightarrow \infty$

T or (F) $f(x) = -x^{100}$ is opening downwards ✓
 so as $x \rightarrow \infty, y \rightarrow -\infty$ and as $x \rightarrow -\infty, y \rightarrow -\infty$

8. What is the difference between the average rate of change and the instantaneous rate of change? Provide a diagram(s) to justify your reasoning.

(2) Average rate of change is the overall rate of change through an interval of time. It is the change between two points of a function, which is the slope of a secant line.

Instantaneous rate of change is the change at one instance in time. It is the change of one point of a function, which is the slope of the tangent line



9. Given the following table determine the equation of the function.

x	y
0	-240
1	-30
2	0
3	0
4	0
5	-90
6	-480

1st 2nd 3rd 4th

$$\begin{array}{l} 210 \\ 30 \\ 0 \\ 0 \\ -90 \\ -390 \end{array} \begin{array}{l} -150 \\ -30 \\ 0 \\ 0 \\ -90 \\ -300 \end{array} \begin{array}{l} 150 \\ -120 \\ 30 \\ -120 \\ -90 \\ -210 \end{array} \begin{array}{l} 45 \\ -120 \\ -120 \\ -120 \\ -120 \\ -300 \end{array}$$

CFD = an!

$$-120 = a(4!)$$

$$a = -5$$

$$y = -5(x-2)(x-3)(x-4)(x-r)$$

$$-240 = -5(0-2)(0-3)(0-4)(0-r)$$

$$-240 = -120r$$

$$r = 2$$

$$\begin{aligned} y &= -5(x-2)(x-3)(x-4)(x-2) \\ &= -5(x-2)^2(x-3)(x-4) \end{aligned}$$

10. Without creating a graph of the function $f(x) = (x-1)(x+4)(x-5)$, state the transformations required to transform $f(x)$ to the function $g(x) = 2x^3 + 14x^2 - 8x - 56$.

$$g(x) = 2x^2(x+7) - 8(x+7)$$

$$= (2x^2 - 8)(x+7)$$

$$= 2(x^2 - 4)(x+7)$$

$$= 2(x+2)(x-2)(x+7)$$

$$= 2(x+2)(x+7)(x-2)$$

$$= 2[(x-1)+3][(x+4)+3][(x-5)+3]$$

$$a = 2$$

$$d = -3$$

• Vertical stretch by a factor of 2

• Horizontal shift left 3 units

11. Determine the equation of the graph on the right given a zero at -6 and 3, and goes through the point (-2, 50).

$$f(x) = a(x+6)(x-3)^2$$

$$f(-2) = a(-2+6)(-2-3)^2$$

$$50 = 100a$$

$$a = \frac{1}{2}$$

$$f(x) = \frac{1}{2}(x+6)(x-3)^2$$

$$f(0) = \frac{1}{2}(0+6)(0-3)^2$$

$$= 27 \Rightarrow \text{above 20 so doesn't work}$$

$$f(x) = a(x+6)(x-3)^4$$

$$f(-2) = a(-2+6)(-2-3)^4$$

$$50 = 250a$$

$$a = \frac{1}{50}$$

$$f(x) = \frac{1}{50}(x+6)(x-3)^4$$

$$f(0) = \frac{1}{50}(0+6)(0-3)^4$$

$$= 9.72 \text{ seems about right on the graph}$$

$$y = \frac{1}{50}(x+6)(x-3)^4$$

